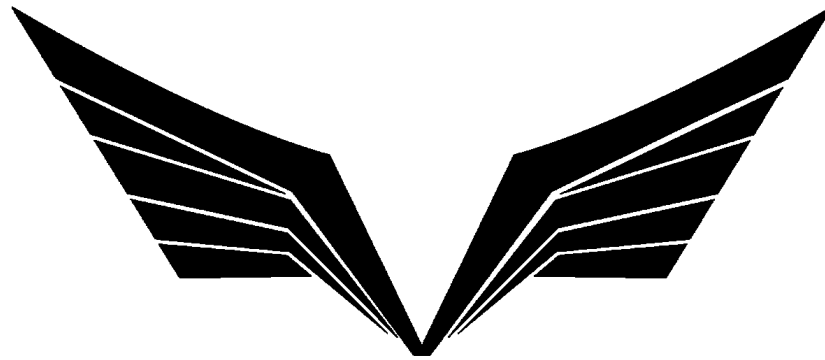




SUN DEVIL

R O C K E T R Y

Avionics Hardware Development Team

New Liquids Feed System (NLFS) System Requirements Document

Project:	New Liquids Architecture
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1. OVERVIEW:

The previous liquid avionics platform required two expensive, custom PCBs as well as all of the hardware and software work needed to build and test them. Every change to a sensor required a full rebuild of the hardware, and every rebuild of the hardware required firmware to be brought back up and tested. There were also significant reliability and maintainability issues on this platform.

So, at the conclusion of the 2025-26 development cycle, the Avionics group at Sun Devil Rocketry has set out to replace this aging system. In particular, the group is looking for a solution to the following pitfalls of last generation:

1. Long hardware lead times
2. Monolithic hardware + firmware components
3. Slow data logging performance
4. Unreliable hardware connections

The solution the group has come up with is a distributed system with a COTS data acquisition and control device at its core.

2. HARDWARE REQUIREMENTS

- 2.1: The system shall be designed to support a liquid engine rocket.
 - 2.1.1: The system shall use the MAV and its extensions when data must be collected onboard the rocket.
 - 2.1.2: The system shall use consumer off the shelf products to collect other data acquisition and control peripherals.
- 2.2: The system shall integrate the LabJack T7.
 - 2.2.1: The LabJack T7 shall be used to collect data.
 - 2.2.2: The LabJackT7 shall be used to provide control inputs.
- 2.3: The system shall have its hardware inputs and outputs labeled.
- 2.4: The system shall have an external “killswitch” that is functional at a safe distance.
 - 2.4.1: The system shall be able to shut off the engine.
 - 2.4.2: The system shall be able to shut off when a mechanical failure occurs.
- 2.5: The system shall produce multiple consecutive sources of telemetry data.
 - 2.5.1: The system shall feed onboard data to the flight computer via CAN.
 - 2.5.2: The system shall be able to collect readings from pressure transducers through the LabJack.
 - 2.5.3: The system shall be able to collect readings from thermocouples through the LabJack.
 - 2.5.4: The system shall be able to log data at a minimum rate of 200 Hz.
- 2.6: The system shall have redundant methods of data storage for data collected in flight.
 - 2.6.1: An external flash chip shall be used to store data onboard.
 - 2.6.2: Data shall be transmitted through LoRa.
- 2.7: The system shall have connections for every component that requires one.
 - 2.7.1: The system shall use “quick disconnects” for check valves during launch.

3. SOFTWARE REQUIREMENTS:

- 3.1. The system shall be written in a high-level programming language.
- 3.1.1. The chosen language shall support an “exception” or “try/catch” construct for graceful error handling.
- 3.2. The system shall feature a graphical user interface as the primary mode of operation.
- 3.2.1. The interface shall display a simplified P&ID containing all inputs (sensor readings) and outputs (control commands) available to the system.
- 3.2.2. The interface shall use color to convey information about the system status in accordance with the table below:

Reading/Control	Color	Meaning
<u>Control</u>	Green	The control interface is open.
	Red	The control interface is closed.
<u>Reading</u>	Green	The sensor reading is within the “normal” range, or the connection is open.
	Orange	The sensor reading is within the “warning” range, or the connection is unstable.
	Red	The sensor reading is outside both the “normal” and “warning” ranges, or the connection has disconnected.

Table 1: Global Interface Color-Coding Scheme

- 3.2.2.1: The colors specified in the Global Interface Color-Coding Scheme shall ONLY be used in the interface for the specified purposes.
- 3.2.3. The interface shall provide a large “**ABORT**” option that sends control inputs to immediately return the system to a safe state.
- 3.2.3.1. If there are “warning” level system messages, the “**ABORT**” option shall be highlighted in orange.
- 3.2.3.2. Otherwise, if the system is not in a safe state, the “**ABORT**” option shall be highlighted in yellow.

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- 3.2.4. The interface shall provide some immediate feedback to all control inputs, even if processing of the command is delayed.
 - 3.2.4.1. If processing of a control input is delayed, the system shall provide feedback when processing of that input is completed.
 - 3.2.5. The interface shall provide settings to connect to the hardware and SDEC-API.
 - 3.2.6. The interface shall display the current status of the connection with SDEC-API and the LabJack.
 - 3.2.7. The interface shall display a rolling chart of the measured chamber pressure and impulse over time.
 - 3.2.8. The interface shall display all of the following derived readings:
 - 3.2.8.1. Mass flow rate (both propellants)
 - 3.2.8.2. Mixture ratio O/F
 - 3.2.8.3. Total impulse (load cells OR estimate for flight capable deployments)
 - 3.2.9. The interface shall display an alert for each of the following conditions:
 - 3.2.9.1. Off-nominal readings (outside warning and normal ranges)
 - 3.2.9.2. Tank pressure below LOX saturation
 - 3.3. The system shall support a method to interface with SDEC-API.
 - 3.3.1. The system shall support simultaneous operations with other API clients.
Note: See [SDEC-API#15](#)
 - 3.3.2. The system shall push new data to connected UI clients at 60 hz or less.
 - 3.3.3. The system shall support operation without SDEC-API connected.
 - 3.3.4. The system shall handle unexpected SDEC-API disconnects without impacting the rest of the system.
 - 3.4. The system shall be able to log data from the LabJack.
 - 3.4.1. The logging system shall be uncoupled from the interface (e.g. support a higher logging rate than what is displayed)
 - 3.5. The system shall support a debug console for logging of events in the system.
 - 3.6. The system must satisfy the following ease of use requirements:
 - 3.6.1. The UI shall be deployable on each client with one click.
 - 3.6.2. The UI shall not require any installation after downloading.
 - 3.6.3. The backend shall be deployable with one command.
 - 3.6.4. The backend shall be installable with one command.

4. REVISION HISTORY:

1.0 - Initial Revision